

Ethlas: Building a highly scalable, interactive Web3 gaming ecosystem with Vertex AI

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With the flexibility
AI and scalability
Cloud, Ethlas is ir
faster and buildir
products so that
can enjoy the bes
experience on its



About Ethlas

Ethlas is a Web3 gaming company on a mission to engineer the future of gaming by unlocking innovative and fresh user experiences. The company developed Battle Showdown, a multiplayer shooter game that enables players to build their own in-game assets. Backed by Web2, Web3, and gaming venture capital funds, Ethlas has a goal of making Web3 accessible and safe for everyone.

Industries: Technology, Games

Google Cloud results

- Achieves 100% increase in inference productivity
- Reduces inference run time from 45 minutes to 10 minutes with Cloud GPUs
- Enables the optimal user experience with Cloud Load Balancing
- Speeds up data analysis to make more informed decisions and respond quickly to changing business requirements

Increase of ML training

Location: Singapore

Load Balancing

Google Cloud (<https://cloud.google.com/>),

Cloud Storage

Cloud Endpoints
(<https://cloud.google.com/endpoints/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Cloud GPUs (<https://cloud.google.com/gpu/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Mobile gaming has taken off in recent years, with the global mobile market worth \$184.4 billion in 2022

(<https://www.precedenceresearch.com/mobile-gaming-market>)

. This number is expected to reach \$775.69 billion by 2032, making it one of the most lucrative industries in the world. Capitalizing on this growth of mobile gaming, Ethlas (<https://ethlas.com/>) developed Battle Showdown, a mobile app that leverages decentralized technologies like blockchain to forge a transparent, secure user-centric gaming experience. In Battle Showdown, gamers are empowered to build their own in-game assets, making every game a unique and enjoyable experience.

"If someone was holding a birthday party, they could create an asset for this party and share it with their friends. Or, if they're celebrating a teacher, they could choose to put their teacher's face on a cute character within the game," explains Elston Sam, Co-Founder and Chief Product Officer at Ethlas.

Having closed its early seed round in January 2022, the team evaluated different cloud providers and decided on Google Cloud (<https://cloud.google.com/>) for its flexibility and scalability. "Our focus was on developing the game and its capabilities using generative AI," says Wui Ngiap Foo, CEO and Co-Founder at Ethlas. "We found that Google Cloud offers the best services particularly in its AI capabilities and went straight for it because we didn't want to have to migrate across vendors later on due to the large-sized files involved. This allows

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Unleashing unlimited creativity with Vertex AI

As a forerunner in the gaming space that lets users take charge of their own in-game assets, Ethlas was an early adopter of Vertex AI (<https://cloud.google.com/vertex-ai>). "We tried Vertex AI while it was still in private preview, and found it to be a great tool because we could have user-managed workflows that allowed for quick deployment and easy setup," says Sam. He adds that the support that the Google Cloud team provided really helped Ethlas expand its use-cases as they experimented with the possibilities of Vertex AI. "Everytime we come up with a use case, the Google Cloud team would present us with a few different options to play around with, and we could discuss the different solutions with the team openly."

Ethlas first builds up a training pipeline, and after curating a training dataset, Ethlas taps on Vertex AI workbench tools to fine-tune stable diffusion models to ensure that the output, such as the art style, is in line with the style of Battle Showdown. A query is then performed for a new output to be delivered. Ethlas leverages Cloud Endpoints (<https://cloud.google.com/endpoints>) to deploy its trained models as APIs for inference to be integrated into its applications.

"Vertex AI's fully-managed service streamlined our deployment process, enabling us to leverage the final inference endpoint more effectively for our specific use case. This resulted in reduced inference times and an overall enhanced user experience when players are using GenAI to customize in-game assets in Battle Showdown," explains Neaton Ang, Data Scientist at Ethlas.

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Hosting and managing large AI models effectively

One of the challenges the company faced was ensuring that it had enough storage space for hosting and managing its large AI models effectively. As AI models require significant computational power, the team faced the challenge of getting enough GPU resources from its initial on-premise infrastructure to efficiently train and deploy its models. This limited its capabilities and slowed down AI development.

With Cloud GPUs (<https://cloud.google.com/gpu/>), the team was able to access GPU resources on-demand, which accelerated training and inference processes for its AI models. "By harnessing the power of cloud GPUs and running training pipelines in parallel, we've achieved a significant reduction in our model training time. This breakthrough has enabled us to

continuously refine our models and explore innovative ideas at a more efficient rate," shares Ang.

Improving workload distribution across resources

Another challenge Ethlas faced initially was the absence of a load balancing mechanism to distribute its workloads. This resulted in uneven resource utilization and slower processing times, which in turn impacted the overall performance and responsiveness of its AI applications.

Leveraging Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>), the company can ensure that every player gets the most seamless user experience, which is crucial for the startup.

"Load balancing is so important to our business especially when it comes to the new user experience," says Sam. "You get one shot. If something doesn't work well, they are unlikely to come back again," says Ang.

He adds that with plans to scale up, load balancing will continue to be a crucial aspect to the business.

"The next step for us is how to ensure that this foundation we've built will cover anything from 1000 users, to 100,000, even 1 million users and beyond."



Interconnectivity across the Google Cloud ecosystem

Ultimately, it is the interconnectedness of the Google Cloud that makes it stand out from competitors.

Ethlas stores its repository of images in Cloud Storage (<https://cloud.google.com/storage>), and can easily connect the relevant buckets with Vertex AI workbench making the implementation of training pipelines very seamless and accessible. Meanwhile, Compute Engine (<https://cloud.google.com/compute>) supports various computing needs through its scalable and customizable VMs.

"Google Cloud has provided us with a flexible and scalable environment for experimentation. We can quickly provision and deprovision resources as needed, enabling us to experiment with different configurations, hyperparameters, and model architectures efficiently so we can make more informed decisions," says Ang.

Moving forward with a trusted ally

Apart from its solutions, Sam believes that a strong partnership and support from the cloud provider is required for a business to truly succeed. "The dedication and unwavering commitment we've seen from the Google Cloud team has been phenomenal," he says. "They are just as passionate as we are about our business and the world of Web3, and their innovative problem-solving skills have been highly regarded within our organization."

Having found a team of like-minded individuals to work with, Sam says, "We look forward to expanding and evolving our partnership with Google Cloud to support our vision of providing AI services, not only internally but also as a consumer-facing external service for our game users."

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